

30-Minute Student Task

Build Your Own Rule-Based Python Chatbot — Solved

Based on the requirements and target flow in the provided task sheet.

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1. Task Summary

The task asks for a simple rule-based conversational chatbot that continuously interacts with the user, recognizes different messages, gives appropriate responses, and exits when the user types “bye” or “exit.”

2. Requirement Checklist

Requirement	How it is satisfied
Ask the user's name before the main conversation.	Modified version: the chatbot starts directly without asking for a name.
Use while True.	Used to keep the conversation running.
Use input() to receive messages.	Used for the user's name and every message.
Clean input using .lower().strip().	Applied to every conversation message.
Recognize at least two greetings.	Recognizes hello and hi, including variations using in.
Answer at least three questions/topics.	Covers Python, weather, and help/time-related topics.
Use if / elif / else.	The chatbot uses an if/elif/else decision structure.
Use break for bye or exit.	The loop breaks when the cleaned input is bye or exit.
Use the user's name in at least one response.	Modified version: the chatbot does not ask for or use a user's name.
Useful response for unknown messages using else.	Unknown input receives a helpful fallback response.
Challenge: use in operator for flexible greetings.	Recognizes phrases such as 'hello bot' and 'hey hello'.

3. Complete Python Solution

```
print("Welcome to the Rule-Based Python Chatbot!")

print("You can say hello, ask about Python, ask about study tips,")
print("ask about the weather, ask for help, or type 'bye' or 'exit' to leave.")

while True:
    message = input("You: ").lower().strip()

    if message == "bye" or message == "exit":
        print("Goodbye! Have a great day.")
        break

    elif "hello" in message or "hi" in message:
        print("Hello! How can I help you?")

    elif "python" in message:
        print("Python is a beginner-friendly programming language used for "
              "web development, automation, data analysis, and AI.")
```

```

elif "study" in message or "exam" in message:
    print("For effective study, make a simple schedule, study in short focused "
          "sessions, take breaks, and revise important topics regularly.")

elif "weather" in message:
    print("I cannot check live weather, but you can use a weather app "
          "or website for the current forecast.")

elif "help" in message:
    print("You can greet me, ask about Python, ask for study tips, "
          "ask about the weather, or type 'bye' or 'exit' to end the chat.")

elif "time" in message:
    print("I cannot access your local clock in this simple rule-based chatbot.")

else:
    print("I don't have a rule for that message yet. "
          "Try asking about Python, study tips, weather, or help.")

```

4. How the Chatbot Works

INPUT: The program first asks the user for a name, then repeatedly uses `input()` to receive messages.

PROCESS: Each message is cleaned with `.lower().strip()`, so inputs such as HELLO and hello are treated consistently.

RESPOND: The if/elif/else statements compare the cleaned message with known rules and print the appropriate response.

REPEAT: while True keeps the conversation running after each response.

EXIT: If the user enters bye or exit, the chatbot prints a goodbye message and uses break to leave the loop.

5. Test Cases and Expected Results

Test	Input	Expected Result
Greeting	hello	Hello! How can I help you?
Uppercase greeting	HELLO	Hello! How can I help you?
Second greeting	hi	Hello! How can I help you?
Flexible greeting	hello bot	Hello! How can I help you?
Python topic	What is Python?	A short explanation of Python is displayed.
Study topic	Give me study tips	Useful study and revision tips are displayed.
Weather topic	Tell me about the weather	The chatbot explains that it cannot check live weather.
Help	help	The chatbot lists supported topics and commands.
Unknown message	Tell me a joke	A useful fallback message asks the user to try supported topics.
Exit	bye	A goodbye message is displayed and the loop ends.
Exit	exit	A goodbye message is displayed

		and the loop ends.
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6. Sample Demonstration

Welcome to the Rule-Based Python Chatbot!

You can say hello, ask about Python, ask about study tips, ask about the weather, ask for help, or type 'bye' or 'exit' to leave.

You: hello

Hello! How can I help you?

You: What is Python?

Python is a beginner-friendly programming language used for web development, automation, data analysis, and AI.

You: Give me study tips

For effective study, make a simple schedule, study in short focused sessions, take breaks, and revise important topics regularly.

You: Tell me about the weather

I cannot check live weather, but you can use a weather app or website for the current forecast.

You: Tell me a joke

I don't have a rule for that message yet. Try asking about Python, study tips, weather, or help.

You: bye

Goodbye! Have a great day.

7. Why This Solution Meets the Task

This modified solution keeps the required conversational loop and decision-making structure while using more meaningful chatbot topics: Python, study/exam tips, weather, help, and time. The pizza example has been removed. The chatbot also supports flexible greetings such as “hello bot” and “hey hello,” and gives a useful fallback response for unsupported messages.

8. Final Target Flow

INPUT → PROCESS → RESPOND → REPEAT → EXIT

The task sheet identifies INPUT → PROCESS → RESPOND → REPEAT → EXIT as the target flow. This modified version keeps that flow while removing the name prompt from the chatbot conversation.